MATH 2102 Linear Algebra II

Chapter 4
Inner Product Spaces

Department of Mathematics, HKU

Chapter 4: Inner Product Spaces

A Inner product

B More on orthogonality

C Self-adjoint and normal operators

D Isometries and unitary operators

E Positive operators

A1. Setting the scene

We shall soon define something called inner product spaces, which is a particular type of vector spaces.

Throughout this chapter, V and W shall denote finite-dimensional inner product spaces over $\mathbb{F}$, unless the content suggests otherwise.

Moreover, whenever we are working on $\mathbb{F}^n$, we shall be working with the standard inner product (which, again, will soon be defined) unless the content suggests otherwise.

A2. Dot product in $\mathbb{R}^n$

For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we defined the dot product

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

and norm

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

where x_i denotes the i-th entry of $\mathbf{x}$ and similarly for y_i.

This is an example of an inner product, which we shall now define for a general vector space.

A3. Inner product in general vector space

An inner product on a vector space V is a function that takes an ordered pair $(\mathbf{u}, \mathbf{v})$ of vectors to a scalar $\langle \mathbf{u}, \mathbf{v} \rangle$ with the following properties:

- $\langle \mathbf{v}, \mathbf{v} \rangle \geqslant 0$ for all $\mathbf{v} \in V$, equality if and only if $\mathbf{v} = \mathbf{0}$;
- additivity in the first slot, i.e. $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \underline{\langle u, w \rangle + \langle v, w \rangle}$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$;
- homogeneity in the first slot, i.e. $\langle \lambda \mathbf{u}, \mathbf{v} \rangle = \underline{\lambda \langle u, v \rangle}$ for all $\mathbf{u}, \mathbf{v} \in V$ and $\lambda \in \mathbb{F}$;
- conjugate symmetry, i.e. $\langle \mathbf{u}, \mathbf{v} \rangle = \underline{\overline{\langle v, u \rangle}}$ for all $\mathbf{u}, \mathbf{v} \in V$.

A vector space with an inner product is called an inner product space.

A3. Inner product in general vector space

Examples of inner products:

- The standard inner product (or ~~Euclidean inner product~~) of $\mathbb{F}^n$ is defined by

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1\overline{y_1} + x_2\overline{y_2} + \cdots + x_n\overline{y_n}.$$

- On $\mathbb{F}^n$ we may also define $\langle \mathbf{x}, \mathbf{y} \rangle = x_1\overline{y_1} + 2x_2\overline{y_2} + \cdots + nx_n\overline{y_n}$.
- On the set of continuous functions $[-1, 1] \to \mathbb{R}$ we may define

$$\langle f, g \rangle = \int_{-1}^{1} f(x)g(x)\, dx.$$

- On $\mathbb{R}[x]$ we may define $\langle p, q \rangle = \displaystyle\int_{-\infty}^{\infty} p(x)q(x)e^{-x}\, dx$.

A3. Inner product in general vector space

The inner product satisfies the following properties:

For any $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\lambda \in \mathbb{F}$,

(a) $\langle \mathbf{v}, \mathbf{0} \rangle = \langle \mathbf{0}, \mathbf{v} \rangle = 0$;

(b) $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \underline{\langle u, v \rangle + \langle u, w \rangle}$;

(c) $\langle \mathbf{u}, \lambda \mathbf{v} \rangle = \underline{\bar{\lambda} \langle u, v \rangle}$.

A4. Norm

Every inner product on V induces a norm defined by

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

The norm satisfies the following properties:

For any $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$,

(a) $\|\mathbf{v}\| \geq 0$, equality if and only if $\mathbf{v} = \mathbf{0}$;

(b) $\|\lambda \mathbf{v}\| =$ $\underline{\quad \lambda \| v \| \quad}$.

A5. Some terminologies

Consider an inner product space V. We have the following terminologies (all learnt in MATH2101, albeit in the context of $\mathbb{R}^n$):

- Two vectors $\mathbf{u}, \mathbf{v} \in V$ are said to be orthogonal if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.
- A unit vector in V is one whose norm is 1.
- An orthogonal set in V is one in which any two elements are orthogonal.
- An orthonormal set in V is an orthogonal set in which every element is a unit vector.
- For any $U \subseteq V$, the orthogonal complement of U, denoted $U^\perp$, is the set of all vectors in V that are orthogonal to every vector in U.

A6. Results in $\mathbb{R}^n$ that hold in general inner product spaces

Results of a geometric nature

For any $\mathbf{u}, \mathbf{v} \in V$, we have the following.

(a) (Pythagoras' Theorem) If $\langle \mathbf{u}, \mathbf{v} \rangle = 0$, then $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

(b) (Cauchy-Schwarz inequality) $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$, equality if and only if $\{\mathbf{u}, \mathbf{v}\}$ is linearly dependent.

(c) (Triangle inequality) $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$, equality if and only if one of $\mathbf{u}$, $\mathbf{v}$ is a non-negative scalar multiple of the other.

(d) (Parallelogram law) $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$.

A6. Results in $\mathbb{R}^n$ that hold in general inner product spaces

Properties of the orthogonal complement

(a) For any $U \subseteq V$, $U^{\perp}$ is a subspace of V.

(b) $\{\mathbf{0}\}^{\perp} = \underline{\ V\ }$ and $V^{\perp} = \underline{\ \{0\}\ }$.

(c) If $U_1 \subseteq U_2 \subseteq V$, then $U_1^{\perp} \underline{\ \supseteq\ } U_2^{\perp}$.

A6. Results in $\mathbb{R}^n$ that hold in general inner product spaces

Properties of orthogonal and orthonormal sets

(a) Every orthogonal set of non-zero vectors in V is linearly independent. In particular every orthonormal set is linearly independent.

(b) Suppose $\{\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n\}$ is an orthonormal basis of V. Then any $\mathbf{v} \in V$ can be written as

$$\mathbf{v} = \underline{\langle v, e_1 \rangle}\,\mathbf{e}_1 + \underline{\langle v, e_2 \rangle}\,\mathbf{e}_2 + \cdots + \underline{\langle v, e_n \rangle}\,\mathbf{e}_n.$$

In particular, $\|\mathbf{v}\| = \underline{\sqrt{\sum \langle v_i, e_i \rangle^2}}$.

A6. Results in $\mathbb{R}^n$ that hold in general inner product spaces

Gram-Schmidt process

Suppose $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ are linearly independent vectors in V. Define $\mathbf{w}_1 = \mathbf{v}_1$ and

$$\mathbf{w}_k = \mathbf{v}_k - \frac{\langle \mathbf{v}_k, \mathbf{w}_1 \rangle}{\|\mathbf{w}_1\|^2}\mathbf{w}_1 - \cdots - \frac{\langle \mathbf{v}_k, \mathbf{w}_{k-1} \rangle}{\|\mathbf{w}_{k-1}\|^2}\mathbf{w}_{k-1}$$

for $k \in \{2, 3, \ldots, m\}$. Normalise each $\mathbf{w}_i$ by taking $\mathbf{u}_i = \dfrac{1}{\|\mathbf{w}_i\|}\mathbf{w}_i$. Then

$$\mathrm{Span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\} = \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$$

for $k \in \{1, 2, \ldots, m\}$, and that $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ is an orthonormal set.

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B1. Riesz representation theorem

In MATH2101 we learnt that every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be represented by matrix multiplication, i.e. $T(\mathbf{x}) = A\mathbf{x}$ for some $A \in \mathbb{R}^{m\times n}$.

Now let's turn our attention to linear functionals on an inner product space. It turns out that such can be represented by inner product with a fixed vector, as we will prove soon. Can you find such representation for the following examples?

- $f \in (\mathbb{F}^3)'$ defined by $f(x, y, z) = 4x - 5y + 2z$
- $f \in (P_2(\mathbb{R}))'$ defined by $f(p(x)) = \displaystyle\int_{-1}^{1} p(x)\cos(\pi x)\, dx$

B1. Riesz representation theorem

Riesz representation theorem

For every $f \in V'$, there exists a unique $\mathbf{v} \in V$ such that

$$f(\mathbf{u}) = \langle \mathbf{u}, \mathbf{v} \rangle \quad \text{for every } \mathbf{u} \in V.$$

Proof of uniqueness:

Suppose there exist two such $\mathbf{v}$'s; call them $\mathbf{v}_1$ and $\mathbf{v}_2$. That is,

$$f(\mathbf{u}) = \langle \mathbf{u}, \mathbf{v_1} \rangle = \langle \mathbf{u}, \mathbf{v_2} \rangle \quad \text{for every } \mathbf{u} \in V.$$

$\langle u, v_1 \rangle - \langle u, v_2 \rangle = 0$

$\therefore \langle u, v_1 - v_2 \rangle = 0 \quad \forall u \in V$

i.e. $v_1 - v_2 = 0$.

B1. Riesz representation theorem

Riesz representation theorem

For every $f \in V'$, there exists a unique $\mathbf{v} \in V$ such that

$$f(\mathbf{u}) = \langle \mathbf{u}, \mathbf{v} \rangle \quad \text{for every } \mathbf{u} \in V.$$

Proof of existence:

Take an orthonormal basis $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ of V. Define

$$\mathbf{v} = \underline{\langle v, e_1\rangle}\,\mathbf{e}_1 + \underline{\langle v, e_2\rangle}\,\mathbf{e}_2 + \cdots + \underline{\langle v, e_n\rangle}\,\mathbf{e}_n.$$

Then $f(\mathbf{u}) = \langle \mathbf{u}, \mathbf{v} \rangle$ since

$$f(v) = f\left(\sum \langle v, e_i\rangle e_i\right) \to \mathbb{F}$$

$$= \sum \langle v, e_i \rangle f(e_i)$$

B1. Riesz representation theorem

Now we can return to the previous example where $f \in (P_2(\mathbb{R}))'$ is defined by

$$f(p(x)) = \int_{-1}^{1} p(x)\cos(\pi x)\ dx.$$

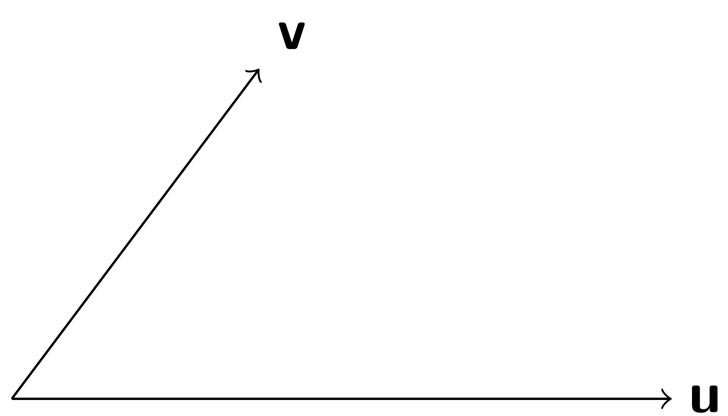
Suppose we adopt the inner product $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\ dx$. Can you find $q \in P_2(\mathbb{R})$ such that $f(p) = \langle p, q\rangle$?

B2. Orthogonal decomposition

We know (at least so in $\mathbb{R}^n$) that given any nonzero vector $\mathbf{u}$, the projection of $\mathbf{v}$ on $\mathbf{u}$ is given by

$$\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2}\mathbf{u}.$$

Every vector can thus be decomposed into the sum of two parts, one parallel to $\mathbf{u}$ and one orthogonal to $\mathbf{u}$.



B2. Orthogonal decomposition

More generally, given a subspace U of V, every vector $\mathbf{v} \in V$ can be decomposed into the sum of two parts, one in U and one in the orthogonal complement of U. We have learnt this fact when $V = \mathbb{R}^n$, and the following is the precise statement in a general inner product space.

Let U be a subspace of V. Then

$$V = U \oplus U^{\perp}.$$

This implies $\dim U^{\perp} =$ __dim V − dim U__ and $(U^{\perp})^{\perp} =$ __U__.

B3. Orthogonal projection

Let U be a subspace of V and $\{\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_m\}$ be an orthonormal basis of U. For each $\mathbf{v} \in V$, by setting

$$\mathbf{u} = \langle \mathbf{v}, \mathbf{e}_1 \rangle \, \mathbf{e}_1 + \langle \mathbf{v}, \mathbf{e}_2 \rangle \, \mathbf{e}_2 + \cdots + \langle \mathbf{v}, \mathbf{e}_m \rangle \, \mathbf{e}_m$$

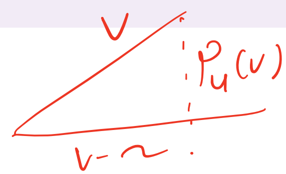
we have $\mathbf{u} \in U$ and $\mathbf{v} - \mathbf{u} \in U^{\perp}$. It is easy to check that the map that sends $\mathbf{v}$ to $\mathbf{u}$ is linear, and is called the orthogonal projection of V onto U, denoted by P_U. Moreover, P_U satisfies the following properties:

- $P_U(\mathbf{v}) =$ __v__ if $\mathbf{v} \in U$ and $P_U(\mathbf{v}) =$ __0__ if $\mathbf{v} \in U^{\perp}$.
- Its kernel is __$U^{\perp}$__ and its image is __U__.
- $P_U^2 =$ __P_U__.

B3. Orthogonal projection

One of the most important properties of orthogonal projection is the following.

> Let U be a subspace of V. Then for any $\mathbf{u} \in U$ and $\mathbf{v} \in V$, we have
>
> $$\|\mathbf{v} - P_U(\mathbf{v})\| \leq \|\mathbf{v} - \mathbf{u}\|,$$
>
> and equality holds if and only if $\mathbf{u} = P_U(\mathbf{v})$.



$v - P_U(v) \perp P_U(v)$

$$\|v - P_U(v)\|^2 \leq \|v - P_U(v)\|^2 + \|P_U(v) - u\|^2$$
$$= \|v-u\|^2.$$

B4. Pseudoinverse

We know that not every linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution. When it is not consistent, we could try to look for the best approximate solution (by finding $\mathbf{x}$ for which $\|A\mathbf{x} - \mathbf{b}\|$ is as small as possible). When a system has infinitely many solutions (or is inconsistent but has infinitely many best approximate solutions), we often want to look for one with the least norm.

The situation is similar for linear transformations. Suppose $T \in \mathcal{L}(V, W)$. If T is bijective, we can define $T^{-1} \in \mathcal{L}(V, W)$. Otherwise, we want a map that sends every vector $\mathbf{w} \in W$ to the 'best approximate solution / solution with least norm' of the equation $T(\mathbf{v}) = \mathbf{w}$.

B4. Pseudoinverse

The following result is key to our work.

Let $T \in \mathcal{L}(V, W)$. Then $T_{(\ker T)^\perp}$ is injective and has the same range as T.

Proof:

To prove injectivity, suppose $T(\mathbf{v}) = \mathbf{0}$ for some $\mathbf{v} \in (\ker T)^\perp$. Then $\mathbf{v} = \mathbf{0}$ since

To prove the other part, it suffices to show that $\operatorname{im} T \subseteq \operatorname{im} T_{(\ker T)^\perp}$. Suppose $\mathbf{w} = \operatorname{im} T$. Then $T(\mathbf{v}) = \mathbf{w}$ for some $\mathbf{v} \in V$.

B4. Pseudoinverse

$TT^{\dagger} = P_{\mathrm{im}\,T}(w)$

$T^{\dagger}T = P_{(\mathrm{null}\,T)^{\perp}}(w)$

Suppose $T \in \mathcal{L}(V, W)$. Define $T^{\dagger} \in \mathcal{L}(W, V)$ by

$$T^{\dagger}(\mathbf{w}) = (T_{(\ker T)^{\perp}})^{-1} P_{\mathrm{im}\,T}(\mathbf{w}).$$

Such $T^{\dagger}$ is called the pseudoinverse of T.

$T^{\dagger}T = (T_{(\ker T)^{\perp}})^{-1} P_{\mathrm{im}\,T} \, T$

Clearly $T^{\dagger} = T^{-1}$ if the latter exists. We state the following property of $T^{\dagger}$, which says it satisfies exactly what we wanted:

For any $\mathbf{v} \in V$ and $\mathbf{w} \in W$,

$$\|T(T^{\dagger}(\mathbf{w})) - \mathbf{w}\| \leq \|T(\mathbf{v}) - \mathbf{w}\|,$$

and equality holds if and only if $\mathbf{v} \in T^{\dagger}(\mathbf{w}) + \ker T$. Moreover, among all such $\mathbf{v}$ that makes equality hold, $T^{\dagger}(\mathbf{w})$ is the unique choice with least norm.

B4. Pseudoinverse

Define $T : \mathbb{F}^3 \to \mathbb{F}^2$ by $T(x, y, z) = (x + 2y + 3z, 0)$. Can you find $T^\dagger$?

$\operatorname{im} T = \operatorname{span}\{(1,0)\}$

$\ker T = \operatorname{span}\{(2,-1,0), (3,0,-1)\}$

$\ker T^\perp = \operatorname{span}\{(1,2,3)\}$

$P_{\operatorname{im} T}(x,y) = (x,0)$

$$T^\dagger = \left(T|_{\ker T^\perp}\right)^{-1} P_{\operatorname{im} T}(x,y)$$

$$= \left(T|_{\ker T^\perp}\right)^{-1}(x,0)$$

$$= \left(\frac{x}{6}, \frac{x}{3}, \frac{x}{2}\right)$$

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C1. Adjoint

Suppose $T \in \mathcal{L}(V, W)$. The adjoint of T is the linear map $T^* : W \to V$ satisfying

$$\langle T(\mathbf{v}), \mathbf{w} \rangle = \langle \mathbf{v}, T^*(\mathbf{w}) \rangle \quad \text{for every } \mathbf{v} \in V \text{ and } \mathbf{w} \in W.$$

Can you explain why T^* is well-defined? What is T^* if $T : \mathbb{R}^3 \to \mathbb{R}^2$ is defined by

$$T(x, y, z) = (y + 3z, 2x)?$$

C1. Adjoint

It is easy to check that the adjoint has the following properties (whenever the operations are well-defined):

$T, S \in \mathcal{L}(V, W)$ $S \in \mathcal{L}(W, U)$

(a) $(S+T)^* = S^* + T^*$

(b) $(\lambda T)^* = \lambda(T^*)$

(c) $(T^*)^* = T$

(d) $(ST)^* = T^* S^*$

(e) $(T^{-1})^* = (T^*)^{-1}$

$I^* = I$

C1. Adjoint

For any $T \in \mathcal{L}(V, W)$, we have

(a) $\ker T^* = (\operatorname{im} T)^\perp$

(b) $\operatorname{im} T^* =$ (range T)$^\perp$

(c) $\ker T =$ (im T*)$^\perp$

(d) $\operatorname{im} T =$ (range T*)$^\perp$

Do these remind you of something that you previously learnt?

C1. Adjoint

Suppose $T \in \mathcal{L}(V, W)$ and fix orthonormal bases for V and W. Then

$$\mathcal{M}(T^*) = (\mathcal{M}(T))^*,$$

where the * on the right refers to the conjugate transpose of a matrix.

$$A_{ij} = \overline{A^*_{j,i}}$$

matrix

C2. Self-adjoint operators

An operator $T \in \mathcal{L}(V)$ is said to be self-adjoint (or Hermitian) if $T^* = T$.

Clearly the following are equivalent conditions for T to be self-adjoint:

- $\langle T(\mathbf{v}), \mathbf{w}\rangle = \langle \mathbf{v}, T(\mathbf{w})\rangle$ for all $\mathbf{v}, \mathbf{w} \in V$.
- The matrix representation of T with respect to some orthonormal basis of V satisfies $(\mathcal{M}(T))^* = \mathcal{M}(T)$, i.e. is a self-adjoint matrix (or a Hermitian matrix).

○ all eigenvalue are real.

○ $\exists A, B$. $AB = BA$ and $T = A + Bi$

C2. Self-adjoint operators

Suppose $T \in \mathcal{L}(V)$.

- If $\mathbb{F} = \mathbb{C}$, then $\langle T(\mathbf{v}), \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in V$ if and only if $T = 0$.
- If $\mathbb{F} = \mathbb{C}$, then $\langle T(\mathbf{v}), \mathbf{v} \rangle \in \mathbb{R}$ for all $\mathbf{v} \in V$ if and only if T is self-adjoint.
- If T is self-adjoint, then $\langle T(\mathbf{v}), \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in V$ if and only if $T = 0$.

如果 $u, w \in V$ on $\mathbb{C}$，那么有

$$\langle Tu, w \rangle = \frac{\langle T(u+w), u+w \rangle - \langle T(u-w), u-w \rangle}{4} + \frac{\langle T(u+\mathrm{i}w), u+\mathrm{i}w \rangle - \langle T(u-\mathrm{i}w), u-\mathrm{i}w \rangle}{4}\mathrm{i}.$$

V on $\mathbb{R}$

$$\langle Tu, w \rangle = \frac{\langle T(u+w), u+w \rangle - \langle T(u-w), u-w \rangle}{4}.$$

C3. Normal operators

An operator $T \in \mathcal{L}(V)$ is said to be normal if $T^*T = TT^*$.

Clearly the following are equivalent conditions for T to be normal:

- $\|T(\mathbf{v})\| = \|T^*(\mathbf{v})\|$ for all $\mathbf{v} \in V$.
- The matrix representation of T with respect to some orthonormal basis of V satisfies $(\mathcal{M}(T))^*\mathcal{M}(T) = \mathcal{M}(T)(\mathcal{M}(T))^*$, i.e. is a normal matrix.

C3. Normal operators

Let T be a normal operator on V. Then

(a) $\ker T^* = \underline{\ker T}$

(b) $\operatorname{im} T^* = \underline{\operatorname{im} T}$

(c) $V = \ker T \underline{\oplus} \operatorname{im} T$

(d) $T - \lambda I$ is normal

(e) $Tv = \lambda v$ iff $T^* v = \bar{\lambda} v$

(f) the eigenvectors wordinate to distinct eigenvalue are $\perp$

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D1. Isometries

An isometry is a linear map which preserves norms. That is, $T \in \mathcal{L}(V, W)$ is said to be an isometry if $\|T(\mathbf{v})\| = \|\mathbf{v}\|$ for all $\mathbf{v} \in V$.

Can you think of some examples, especially those with geometric meaning?

D1. Isometries

Let $T \in \mathcal{L}(V, W)$. The following are equivalent.

(a) T is an isometry.

(b) $T^* T = I$.

(c) $\langle T(\mathbf{v}_1), T(\mathbf{v}_2)\rangle = \langle \mathbf{v}_1, \mathbf{v}_2\rangle$ for all $\mathbf{v}_1, \mathbf{v}_2 \in V$.

(d) T maps an orthonormal set to an orthonormal set.

(e) For any orthonormal bases $\mathcal{B}$ of V and $\mathcal{C}$ of W, the columns of $[T]_{\mathcal{B}\mathcal{C}}$ form an orthonormal set.

It is easy to show that (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d).

D1. Isometries

It remains to show that (d) $\Rightarrow$ (e) $\Rightarrow$ (a).

(d) T maps an orthonormal set to an orthonormal set.

(e) For any orthonormal bases $\mathcal{B}$ of V and $\mathcal{C}$ of W, the columns of $[T]_{\mathcal{B}\mathcal{C}}$ form an orthonormal set.

(a) T is an isometry.

D2. Unitary operators

An operator on V is said to be unitary if it is an invertible isometry. It follows that

Let $T \in \mathcal{L}(V)$. The following are equivalent.

(a) T is unitary.

(b) $T^*T = TT^* = I$, i.e. $T^{-1} = T^*$.

(c) T maps an orthonormal basis of V to an orthonormal basis of V.

(d) For any orthonormal basis $\mathcal{B}$ of V, the columns of $[T]_{\mathcal{B}}$ form an orthonormal set.

(e) T^* is unitary.

(f) infite dim: isometry $\to$ unitary

prop: ① $|\lambda| = 1$

② if $F = \mathbb{C}$, T is unitary iff T's eigenvalue $\to$ V's orghonormal bas and $|\lambda_i| = 1$

D3. Unitary matrices

A matrix $Q \in \mathbb{F}^{n\times n}$ is said to be a unitary matrix (or orthogonal matrix if $\mathbb{F} = \mathbb{R}$) if its columns form an orthonormal basis of $\mathbb{F}^n$. It follows that

Suppose $Q \in \mathbb{F}^{n\times n}$. The following are equivalent.

(a) Q is a unitary matrix.

(b) The ~~rows~~ columns of Q form an orthonormal basis of $\mathbb{F}^n$.

(c) $\|Q\mathbf{v}\| = \|\mathbf{v}\|$ for every $\mathbf{v} \in \mathbb{F}^n$.

(d) $Q^*Q = QQ^* = I$.

D4. QR factorisation

Let A be a square matrix. A QR factorisation (also called QU factorisation/decomposition) of A is a factorisation of A into the form

$$A = QR,$$

where Q is a unitary matrix and R is an upper triangular matrix with positive entries on the diagonal, e.g.

$$\begin{bmatrix} 0 & 4 & 1 \\ 2 & 3 & 5 \\ 0 & 0 & -7 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 5 \\ 0 & 4 & 1 \\ 0 & 0 & 7 \end{bmatrix}.$$

D4. QR factorisation

$a_i = \sum q_k R_{j,k}$

$w_k = a_k - \sum \frac{\langle a_k, w_i \rangle}{\|w_i\|} w_i \qquad q_k = \frac{w_k}{\|w_k\|}$

Every invertible matrix A has a unique QR factorisation.

$A_{i,j} = (a_j)_i =$ $q_1, q_2, \dots q_n$

$Q \qquad R$

Proof of existence:

Using Gram-Schmidt process, turn the columns $\mathbf{a}_1$, $\mathbf{a}_2$, ..., $\mathbf{a}_n$ of A into an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, ..., \mathbf{q}_n\}$ of $\mathbb{F}^n$. Let Q be the matrix whose i-th column is $\mathbf{q}_i$, and R be the matrix whose (i, j)-entry is $\langle a_i, q_j \rangle$. Then

- $A = QR$;
- Q is a unitary matrix.
- R is upper triangular with positive diagonal entries.

D4. QR factorisation

Every invertible matrix A has a unique QR factorisation.

Proof of uniqueness:

Suppose $A = QR = Q'R'$ where Q, Q' are unitary and R, R' are upper triangular with positive diagonal entries.

$a_i, Q_i' \to$ ith column vector of A, Q'

$$a_i = \sum_{j=1}^{n} R'_{j,i} \cdot q_j' = \sum_{j=1}^{i} R'_{j,i} q_j'$$

i.e. $a_i \in \mathrm{span}\{q_1', \ldots, q_i'\}$ → linear independent

equal

$\mathrm{span}\{[q']_i\} = \mathrm{span}\{[q]_i\}$

D4. QR factorisation

As an example, suppose $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -4 \\ 0 & 3 & 2 \end{bmatrix}$. Then $Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{10}} & -\frac{3}{\sqrt{10}} \\ 0 & \frac{3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \end{bmatrix}$ and

7.61 可逆正算子

自伴算子 $T \in \mathcal{L}(V)$ 是可逆正算子，当且仅当 $\langle Tv, v\rangle > 0$ 对任意非零 $v \in V$ 都成立.

7.62 定义：正定（positive definite）

矩阵 $B \in \mathbf{F}^{n,n}$ 称为**正定的**，如果 $B^* = B$ 且

$$\langle Bx, x\rangle > 0$$

对任意非零 $x \in \mathbf{F}^n$ 都成立.

7.63 科列斯基分解（Cholesky factorization）

设 B 是正定矩阵. 那么存在唯一一个对角线上仅含正数的上三角矩阵 R 使得

$$B = R^* R.$$

$\nearrow A = QR.$

$\exists A,\ B = A^* A = R^* Q^* Q R = R^* R$

Chapter 4: Inner Product Spaces

A Inner product
B More on orthogonality
C Self-adjoint and normal operators
D Isometries and unitary operators
E Positive operators

E1. Motivation

Given an operator T on V, we can consider powers of T, or more generally a polynomial of T. How about other operations such as 'taking square root'?

Suppose $T \in \mathcal{L}(V)$. We say that $R \in \mathcal{L}(V)$ is a square root of T if $R^2 = T$.

For example, can you find a square root of the operator $T : \mathbb{F}^3 \to \mathbb{F}^3$ defined by

$$T(x, y, z) = (z, 0, 0)?$$

E2. Definitions and examples

A self-adjoint operator $T \in \mathcal{L}(V)$ is said to be positive if

$$\langle T(\mathbf{v}), \mathbf{v} \rangle \geq 0 \quad \text{for every } \mathbf{v} \in V.$$

Remarks:

- If $\mathbb{F} = \mathbb{C}$, the self-adjoint condition is redundant.
- As we shall see later, it would have been more appropriate to use the name non-negative operator or positive semidefinite operator, although these are much less popular.

E2. Definitions and examples

Are the following operators positive?

(a) (✓/✗) $T : \mathbb{F}^3 \to \mathbb{F}^3$ defined by $T(x, y, z) = (z, 0, 0)$

(b) (✓/✗) $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x, y) = (2x - y, y - x)$

(c) (✓/✗) $P_U \in \mathcal{L}(V)$ where U is a subspace of V

$2x^2 - xy + y^2 - xy$

$- 2x^2 + y^2 - 2xy$

≥ 0

E3. Properties

The key property of a positive operator is the following.

Every positive operator T has a unique positive square root, denoted by $\sqrt{T}$.

We shall prove the above result in the next chapter, while assuming it we can deduce the following.

An operator $T \in \mathcal{L}(V)$ is positive if and only if $T = R^*R$ for some $R \in \mathcal{L}(V)$.

E3. Properties

A self-adjoint operator $T \in \mathcal{L}(V)$ is positive and invertible if and only if

$$\langle T(\mathbf{v}), \mathbf{v}\rangle > 0 \quad \text{for every nonzero } \mathbf{v} \in V.$$

E4. Definite matrices

A self-adjoint matrix $B \in \mathbb{F}^{n \times n}$ is said to be

- positive semidefinite if $\langle B\mathbf{x}, \mathbf{x} \rangle \geq 0$ for every $\mathbf{x} \in \mathbb{F}^n$;
- positive definite if $\langle B\mathbf{x}, \mathbf{x} \rangle > 0$ for every $\mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$;
- negative semidefinite if $\langle B\mathbf{x}, \mathbf{x} \rangle \leq 0$ for every $\mathbf{x} \in \mathbb{F}^n$;
- negative definite if $\langle B\mathbf{x}, \mathbf{x} \rangle < 0$ for every $\mathbf{x} \in \mathbb{F}^n \setminus \{\mathbf{0}\}$;
- indefinite if it is neither positive semidefinite nor negative semidefinite.

E4. Definite matrices

Classify the following matrices according to their definiteness.

(a) The identity matrix

(b) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

(c) $A^T A$ where A is a real invertible matrix

E5. Cholesky factorisation

For every positive definite matrix B, there exists a unique upper triangular matrix R with positive diagonal entries such that $B = R^*R$.

Proof of existence:

Write $B = A^*A$ and $A = QR$. Then $A^* = R^*Q^*$ and so

E5. Cholesky factorisation

For every positive definite matrix B, there exists a unique upper triangular matrix R with positive diagonal entries such that $B = R^*R$.

Proof of uniqueness:

Suppose $B = R^*R = S^*S$, where R is the matrix so constructed in the proof of existence, and S is an upper triangular matrix with positive diagonal entries. Then $BS^{-1} = S^*$.

E5. Cholesky factorisation

Here is an example of Cholesky factorisation:

$$\begin{bmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 1 & 0 \\ -8 & 5 & 3 \end{bmatrix} \begin{bmatrix} 2 & 6 & -8 \\ 0 & 1 & 5 \\ 0 & 0 & 3 \end{bmatrix}.$$